

CS3 Rubric – Case Study Create

DS 4002 – Spring 2026 - Jackson Glass

Individual Assignment

Submission Format: Upload a link to your github repository on UVA Canvas

Case Study Repository: https://github.com/gzy5jd/DS4002_CS3

Why am I doing this?

This case study provides you with the opportunity to tackle an important real world issue, relating to the study of fake news articles across different topics. In tackling this issue, you will also be working with text-based data, and this case study provides you with an opportunity to learn new skills related to text-based data. In particular, through this case study, you will gain experience working with text vectorization and clustering, through the lens of studying fake and real news.

What am I going to do?

The github repository attached to this assignment provides you with the materials necessary to perform a case study related to the classification of real and fake news stories by topic. The repository contains a folder, Data, which provides instructions for downloading a dataset made up of headlines and article text for real and fake news stories. The specific exploratory analysis performed is up to you, but some recommendations would be title length, article length, and the frequency of symbols such as exclamation marks in real versus fake articles. After this, you will convert the text of news articles into numerical data, using Sci-kit learn's TF-IDF text vectorization implementation. You should explore the output of this method in the code. Finally, you will take the vectorized text, and perform K-means clustering, to attempt to organize the vectorized data into clusters, split by the topic of the news story. Convert the vectorized text back into written text, and try to understand the clustering algorithm.

Now that you have performed modeling, produce a document explaining your findings. This document should give an overview of the analysis procedure, as well as primarily focus on your interpretation of the results. Some guiding questions to consider could include looking at the breakdown of real versus fake news articles in each topic, how you determined the number of topics (clusters) to use, and how you validated the output of these clusters. What sort of information can you get out of the clusters that may be helpful for understanding fake news. This is the most open ended part of the project - try to be creative and come up with interesting ways to interpret the data!

Tips for success:

- Ensure you understand the modeling approaches used before trying to implement them
- When implementing the analysis, try and think of different ways that you can tell if the analysis is successful
- If you don't understand the full model, try to implement a smaller or simpler test version of for your own understanding

- Don't be afraid to try new things, failing the first time prepares you for greater learning and understanding the second time.

How will I know I have Succeeded?

You will know you have succeeded in performing this case study when you are able to meet the specific requirements listed below in the rubric.

Formatting	• Your project should be submitted in the form of a github repository, including the following components: - o SCRIPTS - A folder for all the scripts used to complete the project, with clear file names - o OUTPUT - A folder containing the outputs of the project (plots from exploratory data analysis, a one page pdf on results, and any other output) - o README.md - A file detailing the purpose of the repository, background on the project, and instructions for how to recreate the results - o LICENSE.md - An open source license for the project, explaining permissions and citations. The MIT license is recommended
SCRIPTS	• Produce a folder, SCRIPTS, in the github repository, which holds all the scripts used to complete this project • A Jupyter Notebook should be created for exploratory data analysis and - o This notebook must be well documented, with comments explaining what each part of the code is doing, and markdown comments providing an overview of different sections of the notebook - o File and variable names should be clear, this notebook should be easily readable and interpretable by a reader who has not seen it before
Analysis	• The analysis should include correct implementations of both the TF-IDF method and K-means clustering - o For both methods, you should provide justification for modeling decisions made, such as the number of TF-IDF basis elements, and the number of K-means clusters - o Successful implementation of TF-IDF is achieved by converting the text data into numerical, vectorized text data. - o Successful implementation of K-means is achieved by using sci-kit learn's K-means clustering model, and demonstrating that you have split the original dataset into clusters - o You should implement a function that can be used to qualitatively understand the cluster. This could be something

	like looking at the most common words, but you are free to use whatever method you choose. Interpret the differences between clusters, what do you notice?
Results	● You should produce a text document, approximately one-half page to one page in length, interpreting the results of your analysis ○ This one pager should provide background on the project, explanation of how you implemented the analysis, and what the analysis told you ○ Some of the following questions may be helpful as a guide, but do not have to be followed strictly ▪ How can we analyze the clusters, are we able to interpret a clear topic for each cluster using the above modeling approach? ▪ Can you tell any differences between the clusters? What sort of differences are noticeable (word length, common words, etc.)? ▪ How does this align or not align with any expectations you had? ○ A null result is okay! If your analysis doesn't find significant differences by topic, that is also useful information to have, as long as you explain and justify why.

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